Observation of snow in Antarctica using satellite and *in situ* measurements

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Snow in the climate system

Snow albedo and grain size

Antarctica: study sites

This work: objectives

In situ: Multiband albedometer

In situ: AWS and Solalb

Satellite: S-3/OLCI

Multiband processing

Multiband processing: new implementations

CSI and DIRDIFF

Multiband processing: diagnostics

Diagniostic

S-3/OLCI processing: SICE 1.6

Diagniostic

S-3/OLCI processing: SICE 1.6 diagnostics

Diagniostic

Solalb processing

Diagniostic

SSA time series from Multiband and SICE 1.6

SSA time series from Multiband and SICE 1.6

Multiband diagnostics

Multiband diagnostics

SICE 1.6 diagnostics and Solalb

Geophysical interpretations of the time series

Geophysical interpretations of the time series

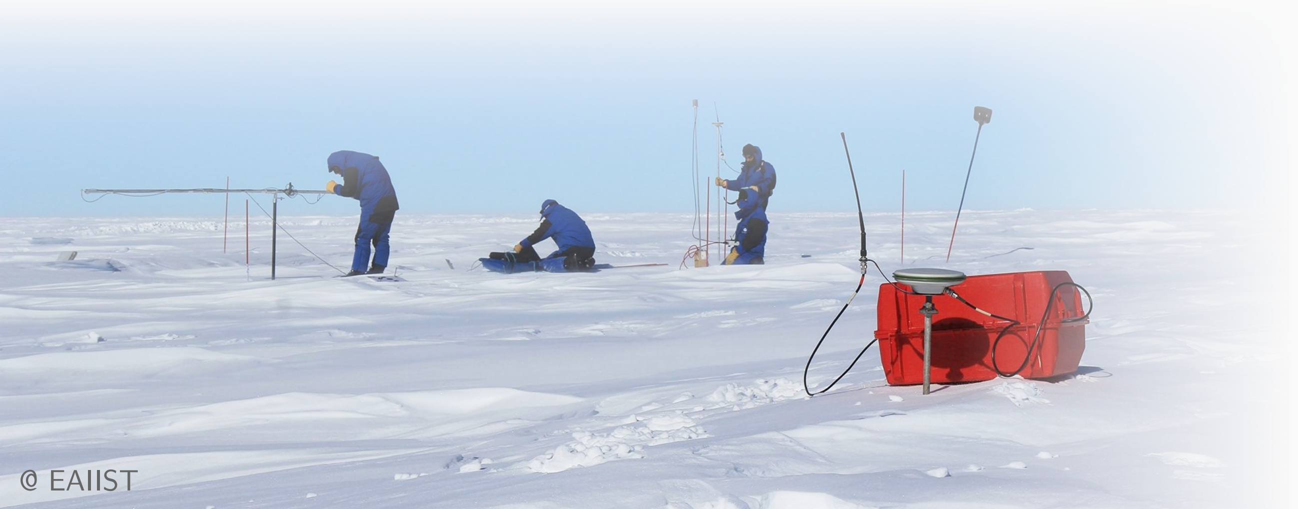
Future implications

Objective of the work

Snow albedo / specific surface area (SSA) time series from *in situ* measurements

Validation of Sentinel-3 albedo estimation

Daily albedo / SSA map for the **whole Antarctic continent**

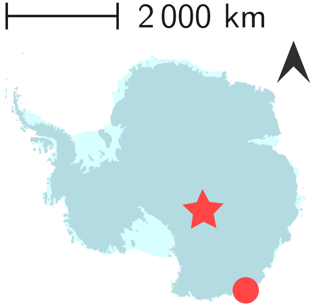


In situ: the Multiband instrument

Multi-spectral-band radiometer
operating in VIS, NIR and SWIR

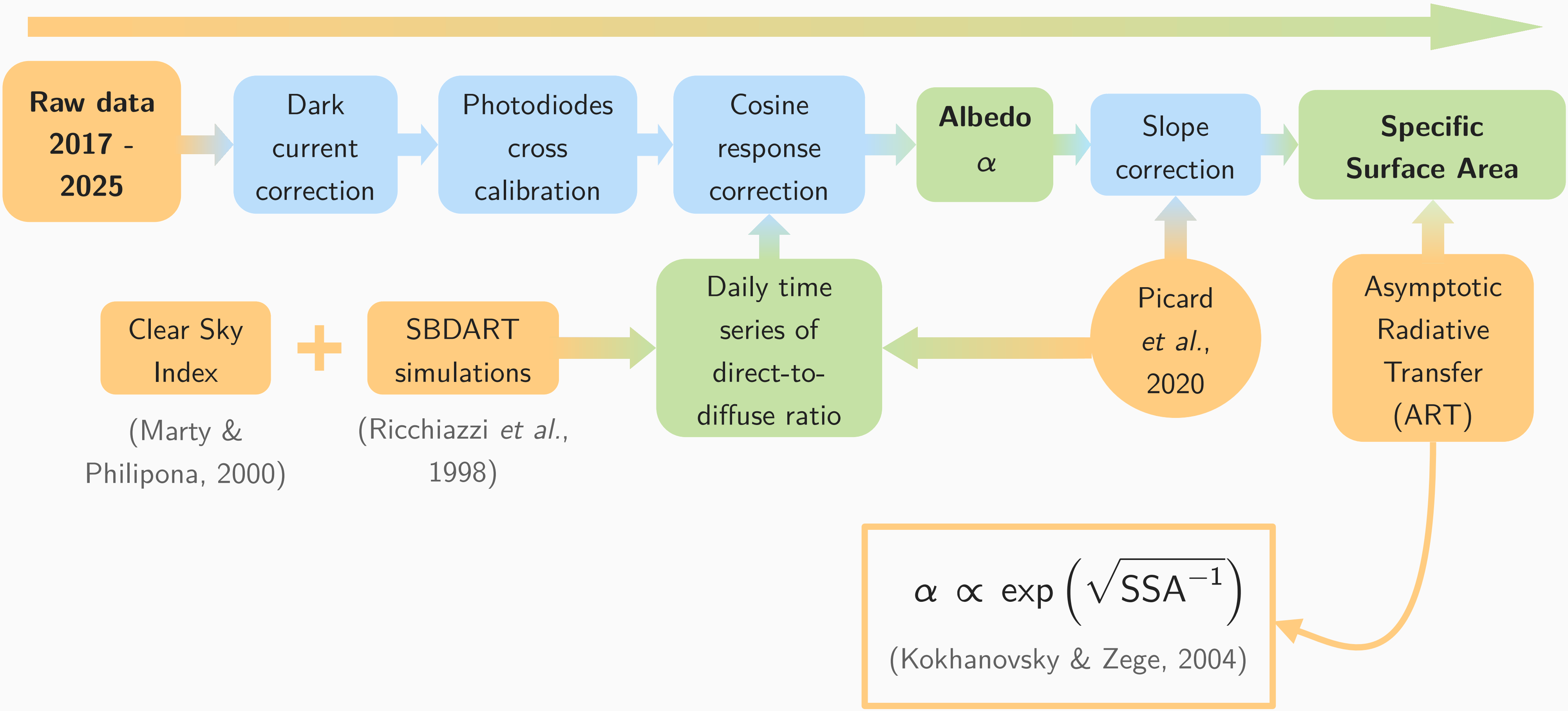
Measures incident and reflected solar radiation → **spectral albedo**

Installed at **Cap Prud'Homme** and in a **megadunes zone** during the **EAIIST** campaign



Picard *et al.*, 2016

Processing of Multiband data (Arioli *et al.*, 2023)



Satellite: Ocean and Land Colour Instrument/Sentinel-3

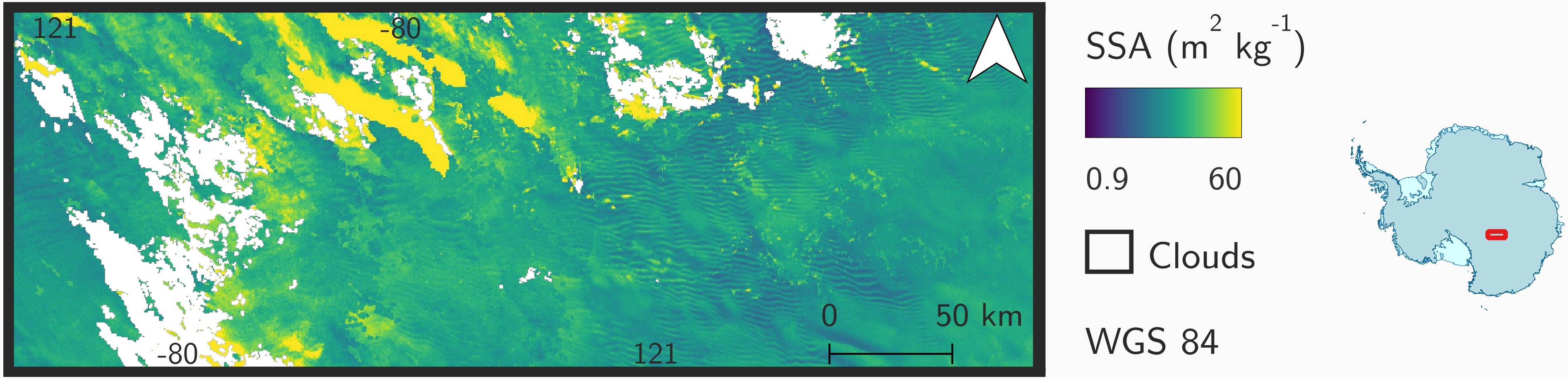
Snow and Ice **SICE 1.6** algorithm for albedo retrieval (Vandecrux *et al.*, 2022)

OLCI/S-3
Band 17: $\lambda_1 = 865$ nm
Band 21: $\lambda_2 = 1020$ nm

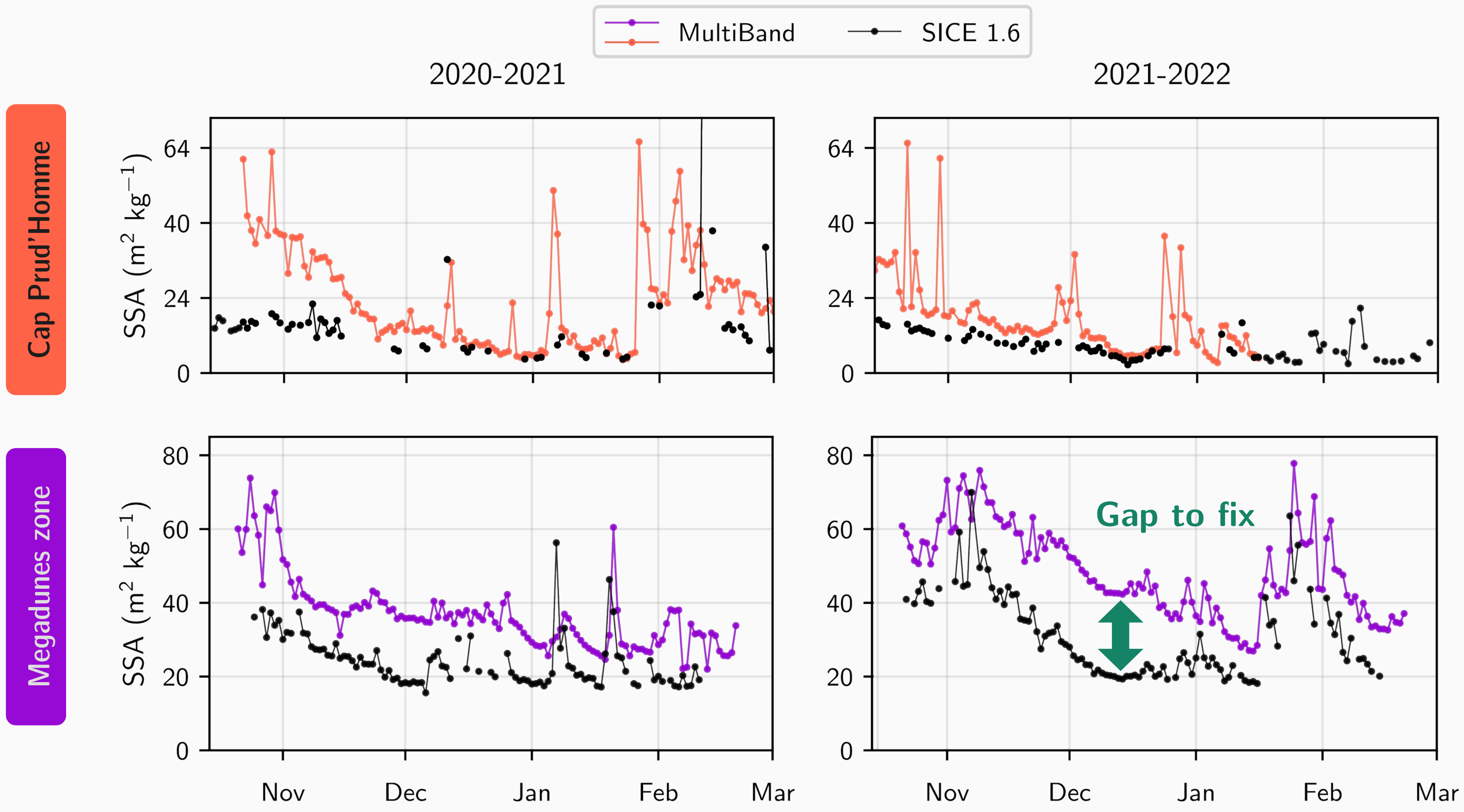
TOA, O_3-corrected
snow reflectance
$R_s(\lambda)$

ART +
Cloud mask
(Poulsen *et al.*, 2020)

Specific
Surface Area



Multiband – SICE 1.6 SSA time series comparison



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Thank you for your attention!